ENGINEERING TECHNICAL DOCUMENTATION

CANscope User & Operations Manual

Standalone CAN / CAN FD Diagnostic Workbench & Telemetry Suite (Release v1.0.0)

1. System Overview & Architecture

CANscope is an air-gapped, license-free desktop diagnostic and telemetry workstation engineered for automotive systems engineers, ECU software developers, and test validation teams. It provides full post-test physical signal decoding, bus timing jitter auditing, functional safety validation, and data interchange without dependency on proprietary commercial vendor software or hardware dongles.

Core Capabilities

- **Direct Trace Ingestion:** Direct parsing of Vector `.blf` (Binary Logging Format) and `.asc` (ASCII Bus Logging) files alongside standard `.dbc` communication matrices.
- **Multi-Trace Oscilloscope:** Multi-channel synchronized analog waveforms and discrete state-machine transition timelines for comparing test traces against reference baselines.
- **Bus Timing & Jitter Profiling:** Automated frame cycle calculation (actual periodicity vs. nominal rates), transmission jitter (Δt), and message drop detection.
- **Functional Safety (ISO 26262):** Automated alive-counter sequence audits (detecting skips, freezes, and resets) and state-machine transition logging.
- **Desktop File System Integration:** Native Windows "Save As" file dialogs for interactive HTML plots, verification test reports, and resampled synchronized CSV exports.

2. System Requirements & Quick Start

CANscope runs as a self-contained 64-bit Windows binary with bundled dependencies.

Parameter	Requirement / Specification
Operating System	Microsoft Windows 10 (64-bit) or Windows 11 (64-bit)
Memory (RAM)	4 GB minimum (8 GB+ recommended for large multi-channel logs)
Disk Storage	~250 MB free disk space
Licensing & Dongles	Zero external license managers, servers, or hardware keys required

Standalone Execution Workflow

1. Download the `CANscope_v1.0.0_win64.zip` distribution from the project repository.
2. Extract the archive into your preferred local folder (e.g., `C:\Tools\CANscope`).
3. Launch `CANscope.exe` directly. No Python runtime installation is required.

3. Data Ingestion & Configuration Setup

Data ingestion is managed via the **Configuration & Setup** sidebar panel on the left of the workspace:

- **Database Definition [.DBC]:** Attach the network communication matrix. This maps raw hexadecimal arbitration IDs to human-readable message names, physical signals, bit starts, lengths, linear conversion factors, and symbolic value tables.
- **Primary Measurement Trace [.BLF / .ASC]:** Attach the target log recorded from the vehicle or HIL test bench. Timestamps and payloads are automatically parsed and indexed.
- **Comparison Traces:** Optionally attach additional trace files to enable synchronized multi-channel comparison in the Graphics window.

4. Workspace Tab Reference

Workspace Tab	Primary Purpose	Operational Capabilities
Graphics [Measurement]	Oscilloscope & Waveforms	Synchronized analog waveforms, categorical step lines, multi-trace overlays, and interactive HTML export.
Bus Statistics	Timing & Jitter Health	Evaluates frame periodicity (min, max, mean, σ), transmission jitter (Δt), and dropped message alerts.
Trace & ASIL Analysis	Functional Safety Audits	ISO 26262 alive-counter sequence audits (skips/freezes/resets) and discrete state transition logs.
Logging & Export Setup	Reporting & Data Interchange	One-click verification test reports (.html) and resampled, time-aligned CSV exports via native Save As dialogs.
Trace Window	CAN / CAN FD Inspector	Chronological raw traffic table with domain-specific payload byte coloring (E2E, UDS/ISO-TP, Standard).

5. Operational Guidance & Best Practices

> **Exporting Synchronized Data:** Before exporting a synchronized CSV in the *Logging & Export Setup* tab, ensure the desired signals are actively selected in the *Graphics [Measurement]* tab. The resampler creates a forward-filled, common time-series matrix across disparate message cycle rates.

All report and file generation actions invoke the native Windows `SaveFileDialog`, allowing you to select your preferred target folder and file name directly.